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EGN-102-003

Mission Concept Proposal

For NASA Mars Research Outpost mission proposal has a team of five. Matthew is our risk and safety lead engineer. He will identify mission risks and develop mitigation strategies. This will include: The risk and safety of all 6 astronauts before travel, during travel, set of plans, actions and control to eliminate potential threats or impacts. Those set plans also include the safety of the astronauts while living on Mars to complete the 5-year mission successfully. Another role he has taken is the habitat & structural engineer. The action for that position includes designing living quarters, workspaces, and structural requirements that implement radiation shielding concepts. The second member of the team is Sophia. Her roles are the life support engineer and the project manager. Her role for the project manager will include coordination of all team activities, ensure subsystem compatibility, and oversee the schedule, and finalizing deliverables. The life support engineer role includes developing systems for air, water, food production, and waste management. Our third member is Meerub. She is our power systems and communication engineer. Those roles of actions include designing electrical power generation, storage systems, evaluating solar, nuclear and hybrid options, developing communication systems, and onboard computing requirements. The fourth member is Aiman. His role must be determined at a later date. Our final member is Daniela. She will focus on transportation and manufacturing. The action of those roles includes Designing cargo delivery concepts, landing systems, surface vehicles, resource utilization, 3D printing, and construction methods.

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NASA needs a long-term research outpost that will successfully support space exploration and technology development on Mars. This mission can provide a critical step toward sustaining life on Mars, access to continuous research and testing our technologies for future and larger missions. The 6 astronauts can provide geology, climate, and even potential past life data. This mission will allow for future developments and groundwork that can be implemented by large-scale space agencies including NASA, SpaceX, and other competing engineering firms. Completing the mission successfully can majorly improve and innovate our current or future technologies. That will also reduce the risk and cost for future missions. That's if Mars' settlement is even feasible. This mission can only be successful if communication over the 5-year span is maintainable, staying within the budget, safely supporting and protecting the 6 astronauts from the Martian environment, and finally designing the outpost for future expansion.

For our initial preliminary design, Let's talk about the location first. Mars does have seasonal time periods. Knowing Mars has seasonal time will greatly impact water accessibility. The seasons depict how the water flows down hillsides and craters. Building our living quarters next to a crater will make water harvesting easier. This is important because it would be best to design structures that are lightweight, inflatable, and radiation protection with Martian regolith. Martian regolith is loose soil and rock covering Mars' surface. The astronauts can use 3D printed structures and implement Martian regolith within the structures. For power systems, solar panels will be utilized. Astronauts can store that energy from solar panels in batteries or any capable storage compacity. Another focus will be creating a renewable recyclable system to create water, air, and waste management. Waste management can also be utilized for plant fertilizers. Life

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support has proposed several factions to counter this constraint. The most developed and viable option will be utilizing controlled environment hydroponic farming. This system would essentially allow us to plant without the availability of non-toxic soil, using water infused with plant nutrients. The plants would have to be inside pressurized and heated chambers due to the Mars cold atmosphere. Our team is working to implement strategies to carefully recycle water being used by the system for 5 years. The astronauts freeze dried feces can also be utilized to grow plants consistently. Designing a greenhouse structure from 3D printing will save a lot on the cost. Another technique used is called Laser Vitreous Multi-Material Transformations. This method uses lasers to melt down the local surface material. After, this clay can be used, and it then solidifies into a strong ceramic structure.

The resources that were used to help create these ideas were 4 books, NASA website post. The first book is called "*Space Ethics*" by *Brian Green*. Which goes in depth on risk, safety, planetary protection and building structures on Mars. The second book is called "*Spacefarers*" by *Christopher Wanjek*. That book expands on Mars as a planet. One of the first websites that helped me understand the elements on Mars was called *NASA: Mars Facts*. Another NASA resource we used is called the *NASA Space Communications* resource. These resources inspired some of the ideas we created as a team to help make our own.

For the mission to be successful, current, upcoming and future engineering problems will need solutions. First would be radiation protection. Using Mars regolith and other materials will provide enough radiation protection to last 5 years. Another problem would be continuous power. Solar is limited by dust storms. That is why having a form of energy storage is important.

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Plus, energy needs to be extremely reliable. Having an active and reliable communication system will greatly impact how everything is managed. Using surface antennas and already existing Mars relay orbiters can help with communication. The materials that would be utilized can vary by strength, capability, protection, weight, and cost. Another challenge is optimizing the mission's efficiency and secure systems while staying within the budget of 50 billion dollars. To ensure that this budget fits our mission needs, we will need to correctly estimate the rough cost of multiple systems and technologies. The most important challenge the astronauts will come across is the food and water. Water is on Mars. The water on Mars is only accessible seasonally. Since we are planning the crew to be there over the span of 5 years, it would be great to equip a rover of the stays near the craters with a drill. Seeing as there's ice on Mars in those craters; we can equip the rover with a drill to harvest this ice. We can then melt the ice into water, treat it with a desalination process, and filter the water through a charcoal mixture to ensure safe drinking fresh water. Astronauts can also accept cargo of oxygen, but there isn't enough time to receive it. Mars will be aligned with Earth every 26 months. The cargo shipment can have an estimated travel time of 7-9 months. Hydrogen peroxide and manganese dioxide can be used to create a small amount of oxygen. That won't be good enough for 5 years. The astronauts can use the Mars atmosphere to create oxygen. Mars is 95% carbon. Using an oxygenator machine will chemically split CO₂. Combine that method with NASA's MOXIE idea, and they will have a chance. Now another method that would be great to create breathable air is electrolysis. That will split the remaining carbon atoms from the oxygen atoms. Communication from Earth to Mars will also be a major factor in the 6 astronauts' survival. Surface antennas will be installed, but to get a signal back between both planets will roughly take 24 minutes. There isn't enough time for

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the astronauts to wait if there was a major issue. To try and solve the problems on the spot, they need on spot computing requirements. That includes systems that continuously go over every working part of the project. For example, error/fault detection, handling, processing, and storing data locally before it is sent out to earth. For cargo landings, the astronauts can use any extra materials from the designed structures and 3D prints to create a landing pad for the cargo. The only difficult part about that will be electronically directing the cargo to the landing pad.

For our project plan, we will have to do regular checks between group members to ensure that their part of the project is within the budget limits of their role. Doing that will help prevent wasted time on research and planning. Making sure each member's job is being done and managed correctly will help become successful on time with each deadline. We are approaching the milestones as if we are astronauts. That gives us a sense of responsibility because hypothetically it's "us" not them. That will help the team make it into our final mission model and more cohesive groupwork.

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